RUNI – Algorithms – Homework 3

Due date: Sunday 30/11/2025

NOTE: the general homework guidelines published in homework 1 are valid for all homework assignments in this course.

Problem 1 (25 pts.):

For each of the following statements, decide whether it is true or false, and prove or refute accordingly.

1. Let $G = (V, E)$ be an **undirected**, and possibly unconnected, graph, and let $a, b \in V$. The number of trees in the forests returned by $DFS(a)$ and $DFS(b)$ is the same.

2. Let $G = (V, E)$ be a **directed** graph and let $a, b \in V$. The numbers of trees in the forests returned by a $DFS(a)$ and $DFS(b)$ is the same.

3. Let G be an **undirected** graph. Assume you run $BFS(s)$. During the run, the vertex v marks the vertex u. In the run of $DFS(s)$ on G, that uses the same adjacency lists as the run of BFS, v will be marked before u.

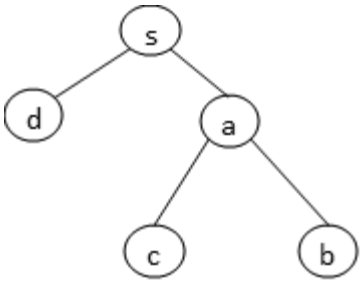
4. Let G be an **undirected** graph that has a Hamiltonian cycle. For every vertex $v \in V$, there exists a run of $DFS(v)$ in which the resulting **directed** spanning tree has a single vertex with an out-degree of 0.

5. Let $G = (V, E)$ be an **undirected** graph with $|V| = 4$. Some run of DFS produces a spanning tree in which the root has degree 2.
   Claim: $|E| \leq 4$.

Problem 2 (15 pts):

1) Let $G_{2,4}$ be the De-Bruijn graph for $n = 4, \Sigma = \{0,1\}$. We perform $BFS(100)$ (namely, starting from the vertex 100). How many vertices will have a label of 2 at the end of the run? Justify your answer.

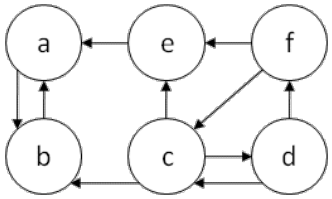
2) Let G be an undirected graph, where $V = \{a, b, c, d, s\}$. $DFS(s)$ produces the following spanning tree.



Draw a graph G that fulfills the above description and has a maximal number of edges. For every $(v, u) \notin E$, justify why this edge cannot be in E.

Problem 3 (20 pts.):

Consider the following directed graph G and let G' denote the underlying graph of G.



Assume that vertices in the adjacency list appear in alphabetic order. For example, c's list is $b \rightarrow d \rightarrow e$.

1) Run $DFS(f)$ on G and draw the resulting spanning tree of G.
2) Run $DFS(f)$ on G' and draw the resulting spanning tree of G'.
3) Run $BFS(d)$ on G and draw the resulting spanning tree of G.
4) Run $BFS(d)$ on G' and draw the resulting spanning tree of G'.

Problem 4 (20 pts.):

A root in a directed graph $G = (V, E)$ is a vertex $r \in V$ such that there is a directed path from r to any other vertex $v \in V$.
Let $G = (V, E)$ be a directed graph. Describe an $O(|V| + |E|)$ time algorithm that finds all the roots in G or states that "a root does not exist". Analyze the running time of your algorithm and explain why it is correct.

Directions: As a first step, use the algorithm for finding a single root in a directed graph (this algorithm was taught in the practical session).

Problem 5 (20 pts.):

Let $G = (V, E)$ be an unweighted, simple directed graph where $s \in V$ is a source vertex and $t \in V$ is a target vertex ($s \neq t$). The graph presents a network of roads. As a result of floods, some of the roads, given by the edge set A, were closed. Suggest an algorithm that decides which road (only one) $e \in A$, should be reopened, so that the shortest path from s to t is as short as possible.

Describe the algorithm formally, explain in several lines its correctness (state the main claim in the proof, but no need to prove it), state and justify the time complexity.

Example: In the given graph, for $A = \{(s, c), (c, b)\}$ if (s, c) is selected to reopen, then the shortest path would be 2. If (c, b) is selected, the shortest path would be 3.

